

Causal Inference Clinics

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Example 1: Power Posing

Background and original claim. The power-posing literature asks whether brief changes in body posture can causally affect psychological, physiological, and behavioral outcomes. The central treatment is whether participants adopt a *high-power pose* or a *low-power pose*. The main outcomes include self-reported feelings of power, hormonal responses, and risk-taking behavior.

The influential original study by Carney, Cuddy, and Yap¹ randomly assigned 42 participants to hold either high-power or low-power poses. Participants in the high-power condition adopted expansive, dominance-related postures; participants in the low-power condition adopted contracted, submissive postures. The study reported that high-power posing increased self-reported feelings of power, increased testosterone, decreased cortisol, and increased financial risk taking. The broader interpretation was that a short, nonverbal behavioral intervention could change internal states and consequential behavior.

Replication design. Ranehill et al.² conducted a conceptual replication with a substantially larger sample of 200 participants. The design followed the original study closely but introduced several changes intended to improve credibility. Participants were randomly assigned to high-power or low-power poses. Posture instructions were delivered by computer, which made the experimenter blind to treatment condition. Participants provided saliva samples before and after the manipulation, allowing the researchers to measure changes in testosterone and cortisol. They also completed incentivized behavioral tasks measuring risk taking in both gain and loss domains, as well as willingness to compete.

The replication therefore preserved the core causal contrast—high-power versus low-power posture—while using a larger sample, a blinded implementation, and additional behavioral measures.

¹Carney, D. R., Cuddy, A. J. C., and Yap, A. J. (2010). Power posing: Brief nonverbal displays affect neuroendocrine levels and risk tolerance. *Psychological Science*, 21(10), 1363–1368.

²Ranehill, E., Dreber, A., Johannesson, M., Leiberg, S., Sul, S., and Weber, R. A. (2015). Assessing the robustness of power posing: No effect on hormones and risk tolerance in a large sample of men and women. *Psychological Science*, 26(5), 653–656.

Findings. Ranehill et al. found that high-power posing increased self-reported feelings of power, which suggests that the manipulation affected participants’ subjective experience. However, they found no significant effect on testosterone, cortisol, financial risk taking, or the additional behavioral tasks. Thus, the subjective feeling-of-power result replicated, but the stronger physiological and behavioral claims did not.

Classroom discussion.

- What exactly is the causal claim: posture affects feelings, hormones, behavior, or all three?
- If the treatment was randomized, why might the original claim still fail?
- Which features of the replication design make the evidence more credible?
- Should we interpret the replication as evidence of no effect, or as evidence that the effect depends on specific conditions?

Example 2: AI Use and Scientific Output

Research question. Hao et al.³ study whether the adoption of artificial intelligence tools changes scientific production and the direction of science. The main independent variable is whether research is *AI-augmented*. The outcomes are measured at two levels: individual scientists’ productivity and impact, and the collective breadth of scientific research.

At the individual level, the outcomes include number of papers, number of citations, and the timing of becoming a research project leader. At the collective level, the outcomes include the breadth of scientific topics studied, the concentration of research attention, and follow-on engagement among papers.

Data. The paper uses a very large bibliometric dataset: 41.3 million research papers from OpenAlex, covering 1980–2025. The analysis focuses on six natural science fields: biology, medicine, chemistry, physics, materials science, and geology. The authors also use Web of Science as a robustness check. The study does not focus on computer science or mathematics, because the goal is to study research that uses AI tools in substantive scientific fields rather than research that develops AI methods themselves.

Measurement of AI use. A central challenge is that AI use is not directly observed. The authors therefore use complex text-analysis tools to identify AI-augmented research. Specifically, they fine-tune a BERT language model using titles and abstracts of papers. The model is trained to classify whether a paper uses AI methods. The classification is validated against expert-labeled data, achieving an F1-score of 0.875. This machine-learning classifier is then applied to the full corpus of papers to construct a paper-level measure of AI use.

³Hao, Q., Xu, F., Li, Y., and Evans, J. (2026). Artificial intelligence tools expand scientists’ impact but contract science’s focus.

Analysis strategy. The authors compare AI-augmented and non-AI papers, as well as AI-adopting and non-AI-adopting researchers, across disciplines and across different eras of AI: machine learning, deep learning, and generative AI. The analysis combines large-scale bibliometric comparisons, citation analysis, career-transition models, and text-embedding methods.

To study the collective direction of science, the authors embed papers into a high-dimensional semantic space using a scientific text-embedding model. They then measure the breadth of knowledge covered by AI and non-AI papers. This allows them to ask whether AI expands or contracts the topical space explored by science.

Findings. The paper reports that AI adoption is associated with substantial individual advantages. Scientists who engage in AI-augmented research publish more papers, receive more citations, and become research project leaders earlier than comparable non-adopters. The headline results are that AI-adopting scientists publish 3.02 times more papers, receive 4.84 times more citations, and become project leaders 1.37 years earlier.

At the same time, the paper finds that AI adoption is associated with a narrowing of collective scientific focus. AI-augmented research covers a smaller topical space, has lower knowledge entropy, and generates less follow-on engagement among related papers. The authors describe this as a tension: AI appears to expand individual scientists' impact while contracting the collective reach of science.

Classroom discussion.

- Is AI causing higher productivity, or are more productive scientists more likely to adopt AI?
- Is AI narrowing science, or is AI more often used in fields that are already data-rich and concentrated?
- Does the big dataset help establish causality?
- Given the data, could you think of a research design that would make the causal claim more credible?